COMPUTER NETWORKS LAB

(Common to CSE, IT and CSE(AI&ML))
III Year – I Semester

Lecture : 3	Internal Marks : 30
Credits : 1.5	External Marks : 70

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Course Objectives

- To learn the connectivity of network devices and implementation of layered protocols.
- To gain hands-on experience with network simulation tools such as Wireshark, Nmap and NS2.

Course Outcomes

Upon successful completion of the course, the students will be able to

- implement datalink and network layer protocols in TCP/IP.
- analyze packet-level data using Wireshark, and apply filters to study specific protocols and traffic behavior.
- perform port scanning and vulnerability detection using Nmap and interpret scan results for security assessment.
- simulate network performance in NS2 to explore packet loss, congestion, throughput and protocol behavior.

List of Experiments:

(Perform any 10 of the following experiments)

1. Study of Network devices in detail and connect the computers in Local Area Network.
2. Write a program to implement the data link layer farming methods such as
   i) Character stuffing ii) Bit stuffing.
3. Write a program to implement data link layer farming method Checksum.
4. Write a program for Hamming code generation for error detection and correction.
5. Write a program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
6. Write a program to implement sliding window protocol for Go back N.
7. Write a program to implement sliding window protocol for Selective Repeat.
8. Write a program to implement Stop and Wait protocol.
9. Write a program for congestion control using Leaky Bucket algorithm
10. Write a program to implement Dijkstra‘s algorithm to compute the shortest path through a graph.
11. Write a program to implement Distance Vector routing algorithm by obtaining routing table at each node (Take an example subnet graph with weights indicating delay between nodes).
12. Write a program to implement Broadcast tree by taking subnet of hosts.
13. Wireshark
    i. Packet capture using Wireshark.
    ii. Starting Wireshark.
    iii. Viewing captured traffic.
    iv. Analysis and statistics & filters.
14. How to run Nmap scan?

15. Operating System Detection using Nmap.
16. Do the following using NS2 Simulator:
    i. NS2 Simulator-Introduction.
    ii. Simulate to find the number of packets dropped.
    iii. Simulate to find the number of packets dropped by TCP/UDP.
    iv. Simulate to find the number of packets dropped due to congestion.
    v. Simulate to compare data rate & throughput.

Reference Books:

1. Computer Networks, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH.

e-Learning Resources:

1. https://www.isi.edu/nsnam/ns/
2. https://www.wireshark.org/
3. https://nmap.org/